



AI,
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PROPOSITIONAL LOGIC

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Business AI Lab

Content

- **Representing problems using logic**
- Propositional logic

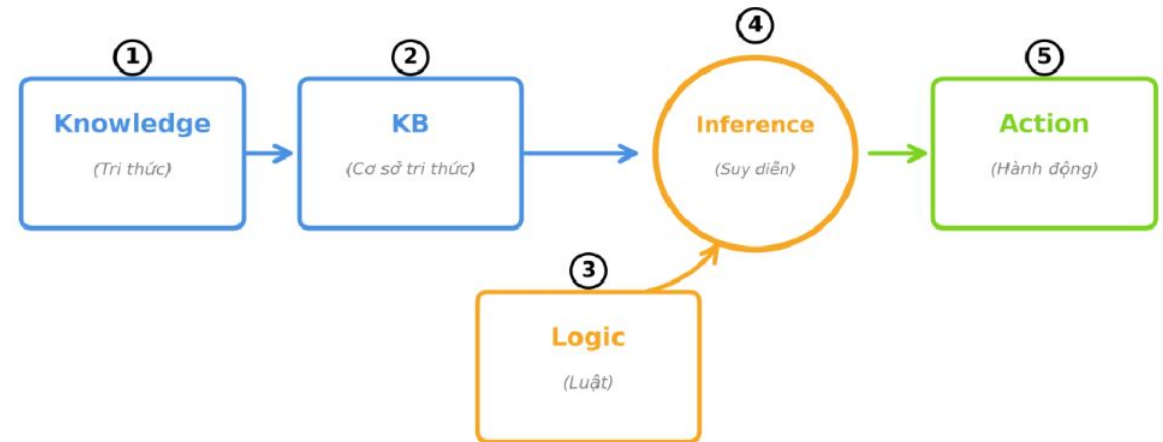
Representing problems using logic

Inference (Reasoning)

- Is the process of **generating a new correct statement** from previously **existing correct statements/ knowledge**, to make appropriate **actions**
 - Knowledge:** Information and knowledge about a certain field. Represented in the form of sentences or propositions
 - Knowledge base (KB): represented in a specific form, creating a knowledge language

Logic is a language in which each sentence in the language represents a **true** or **false value**. Logic:

- Represent the Knowledge
- Reasoning(Inference) based on the Knowledge



Inference Process

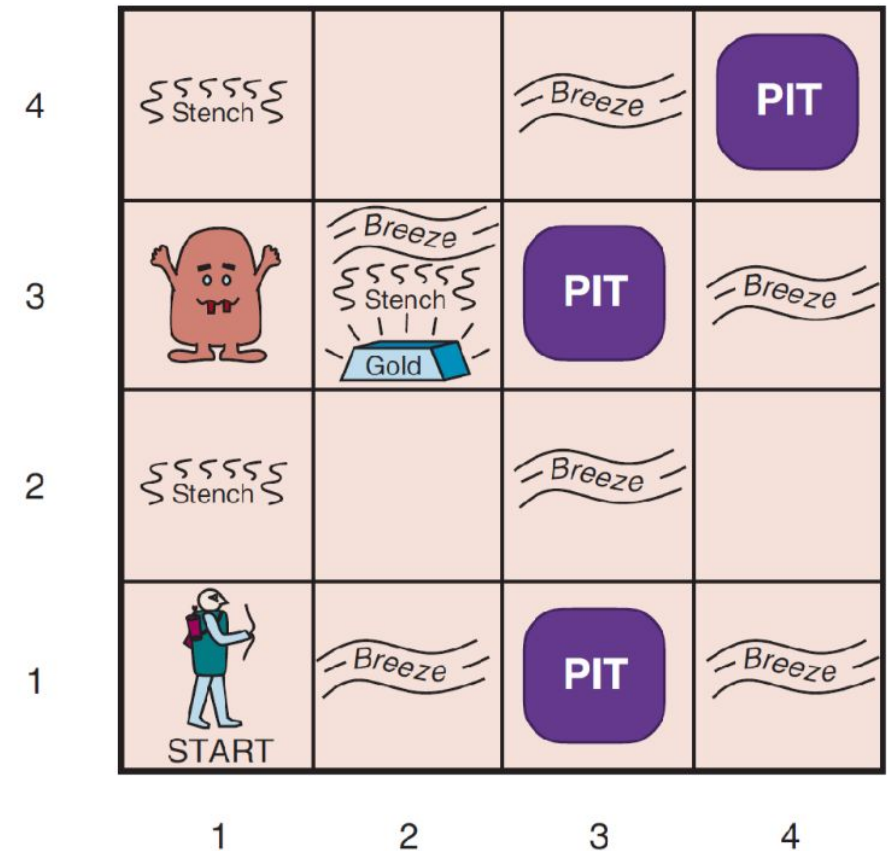
Knowledge → Representation → Reasoning → Action

Representing problems using logic

Problem Description

- A Cave of 4 x 4 rooms
- Wumpus, the monster
- PIT, a bottomless room
- Agent: 5 sensors, 1 arrow,
- Goal: find the Gold and escape from the cave

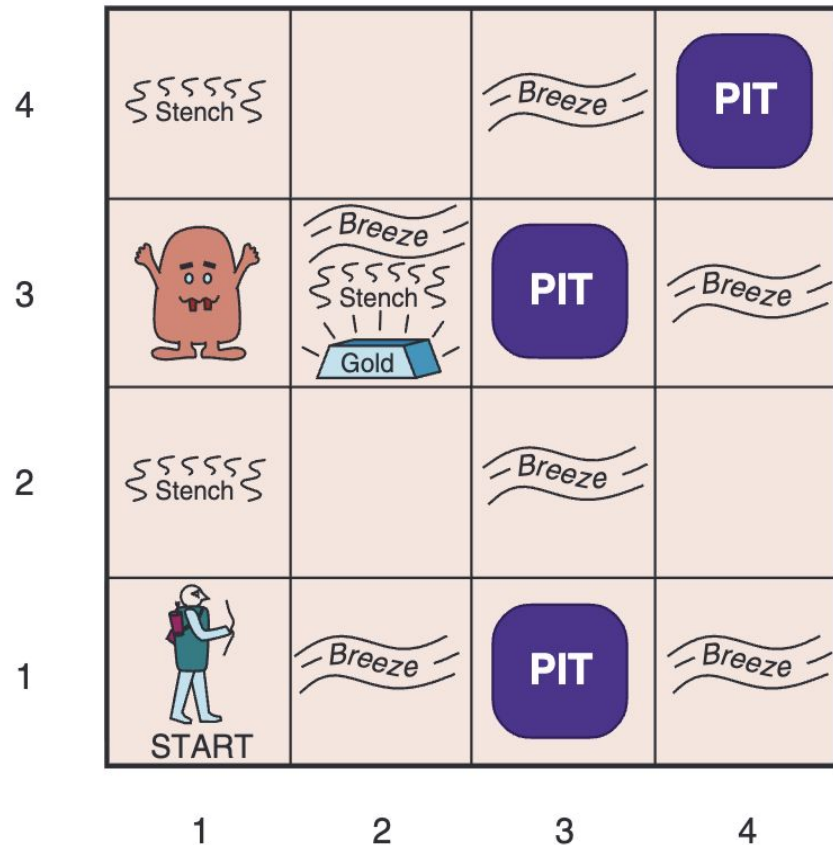
👉 A **Knowledge-Based Agent** is needed



Representing problems using logic

Inference (Reasoning)

PEAS Descriptions



- **Performance measure:** +1000 for climbing out of the cave with the gold, -1000 for falling into a pit or being eaten by the wumpus, -1 for each action taken, and -10 for using up the arrow. The game ends either when the agent dies or when the agent climbs out of the cave.

- **Environment:** A 4×4 grid of rooms, with walls surrounding the grid. The agent always starts in the square labeled [1,1], facing to the east. The locations of the gold and the wumpus are chosen randomly, with a uniform distribution, from the squares other than the start square. In addition, each square other than the start can be a pit, with probability 0.2.
- **Actuators:** The agent can move *Forward*, *TurnLeft* by 90° , or *TurnRight* by 90° . The agent dies a miserable death if it enters a square containing a pit or a live wumpus. (It is safe, albeit smelly, to enter a square with a dead wumpus.) If an agent tries to move forward and bumps into a wall, then the agent does not move. The action *Grab* can be used to pick up the gold if it is in the same square as the agent. The action *Shoot* can be used to fire an arrow in a straight line in the direction the agent is facing. The arrow continues until it either hits (and hence kills) the wumpus or hits a wall. The agent has only one arrow, so only the first *Shoot* action has any effect. Finally, the action *Climb* can be used to climb out of the cave, but only from square [1,1].
- **Sensors:** The agent has five sensors, each of which gives a single bit of information:
 - In the squares directly (not diagonally) adjacent to the wumpus, the agent will perceive a *Stench*.¹
 - In the squares directly adjacent to a pit, the agent will perceive a *Breeze*.
 - In the square where the gold is, the agent will perceive a *Glitter*.
 - When an agent walks into a wall, it will perceive a *Bump*.
 - When the wumpus is killed, it emits a woeful *Scream* that can be perceived anywhere in the cave.

The percepts will be given to the agent program in the form of a list of five symbols; for example, if there is a stench and a breeze, but no glitter, bump, or scream, the agent program will get [*Stench*, *Breeze*, *None*, *None*, *None*].

Representing problems using logic

Inference (Reasoning)

The First Two Steps

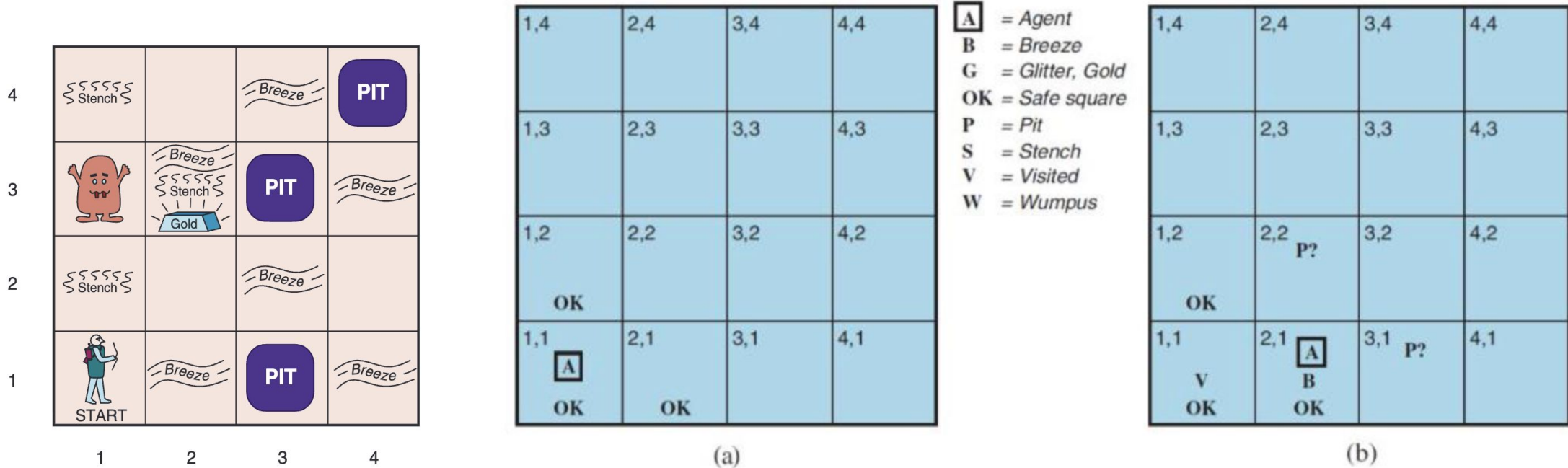
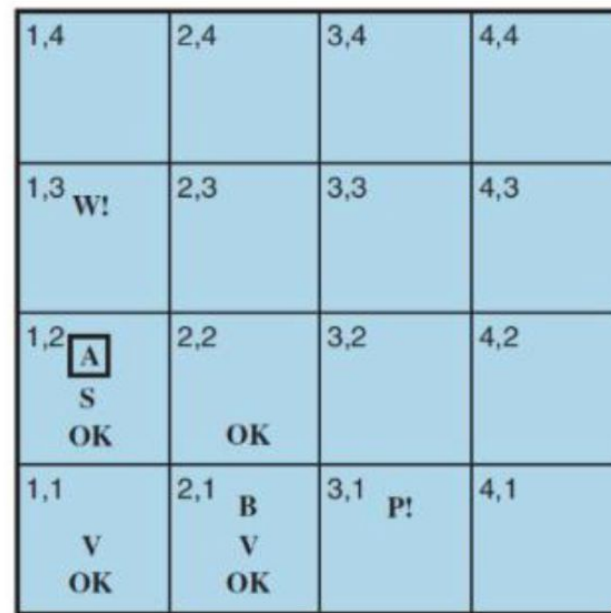
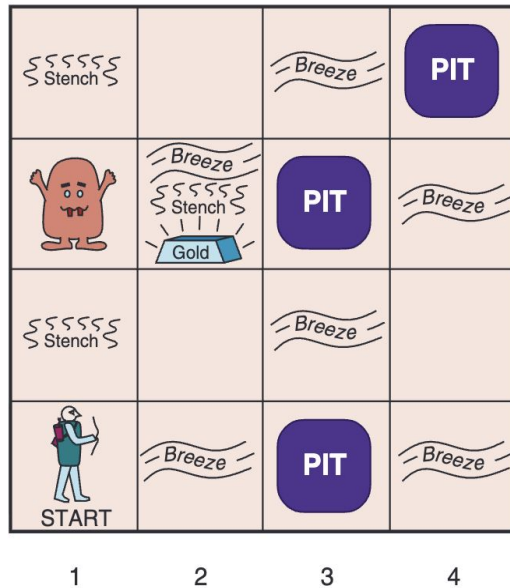


Figure 7.3 The first step taken by the agent in the wumpus world. (a) The initial situation, after percept $[None, None, None, None, None]$. (b) After moving to $[2,1]$ and perceiving $[None, Breeze, None, None, None]$.

Representing problems using logic

Inference (Reasoning)

Two Later Stages



A = Agent
B = Breeze
G = Glitter, Gold
OK = Safe square
P = Pit
S = Stench
V = Visited
W = Wumpus

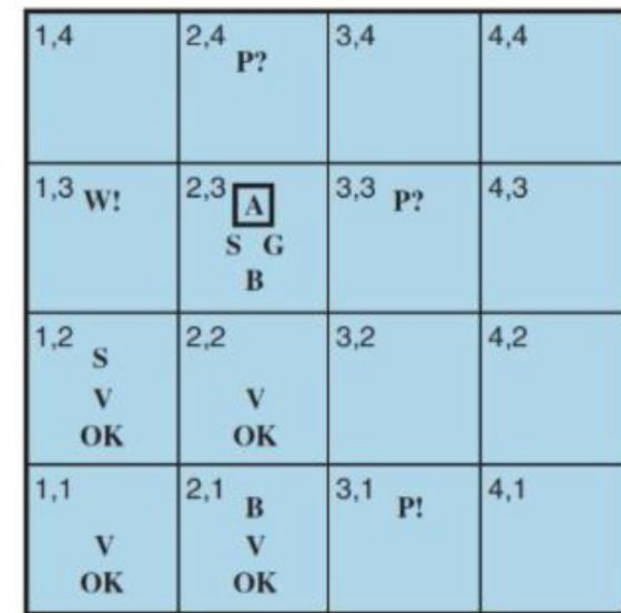


Figure 7.4 Two later stages in the progress of the agent. (a) After moving to [1,1] and then [1,2], and perceiving [Stench, None, None, None, None]. (b) After moving to [2,2] and then [2,3], and perceiving [Stench, Breeze, Glitter, None, None].

Content

- Representing problems using logic
- **Propositional logic**

Propositional logic

Definition

- A proposition is a statement that can have only one of two truth values: **true (T, 1)** or **false (F, 0)**

Reality

"If it rains, the sky is cloudy."

It's raining

So: **"The sky is cloudy"**

Propositional logic

P="it rains"

Q="the sky is cloudy"

Assuming:

- $P \rightarrow Q$: True
- P : True

So $\rightarrow Q$: True (*Modus Ponens rule*)

So: **"The sky is cloudy"**

Propositional logic

Definition

- A proposition is a statement that can have only one of two truth values: **true (T, 1)** or **false (F, 0)**

Reality

"If An had a lot of money, she would go shopping."

An **wouldn't** go shopping

So: "An **don't** have a lot of money"

Propositional logic

P="An had a lot of money"

Q="An go shopping"

Assuming:

- $P \rightarrow Q$: True
- $\neg Q$: True

So $\rightarrow \neg P$: True (*Modus Tollens rule*)

So: "An **don't** have a lot of money"

Propositional logic

Syntax & semantic

- 2 logic constant: True, False
- Parentheses: ()
- Logical operations
 - Conjunction (and): $\wedge$
 - Disjunction (or): $\vee$
 - Negation (not): $\neg$
 - Implication: $\Rightarrow$
 - Equivalence: $\Leftrightarrow$

If P and Q are formulas:

- $P \wedge Q$: P **AND** Q
- $P \vee Q$: P **OR** Q
- $\neg P$: **NOT** P
- $P \Rightarrow Q$: P **Infer** Q
- $P \Leftrightarrow Q$: P **equivalent to** Q

Propositional logic

Syntax & semantic

- **Atomic sentence:** a single clause symbol (or True/False constant). For example: P, Q, Rain.
- **Complex sentence:** combined from smaller sentences using connective. For example: $\neg P$, $P \wedge Q$, $(P \vee Q) \Rightarrow R$.
- **Literal:** an atomic sentence or its negation. For example: P (positive literal), $\neg P$ (negative literal).
- **Clause (clause):** collection ($\vee$) of literals. For example: $(P \vee \neg Q \vee R)$

Atomic Sentences

- An atomic sentence states a fact

Brother(Richard, John) read as "Richard is a Brother of John"

Married(Father(Richard), Mother(John))

- can be *True* or *False* -- semantics/interpretation

```
Sentence      → AtomicSentence | ComplexSentence
AtomicSentence → True | False | P | Q | R | ...
ComplexSentence → ¬ Sentence
                | ( Sentence ∧ Sentence )
                | ( Sentence ∨ Sentence )
                | ( Sentence ⇒ Sentence )
                | ( Sentence ⇔ Sentence )
```

Complex Sentences

- logical connectives($\neg$, $\wedge$, $\vee$, $\Rightarrow$, $\Leftrightarrow$, ...) along with atomic sentences

$\neg \text{Brother(Richard, John)}$

$\neg \text{King(Richard)} \Rightarrow \text{King(John)}$

Propositional logic

Syntax & semantic

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
false	false	true	false	false	true	true
false	true	true	false	true	true	false
true	false	false	false	true	false	false
true	true	false	true	true	true	true

Truth table of logical operations

Propositional logic

Syntax & semantic

- These logical operations have the properties of **commutativity**, **associativity**, and **distributivity**.

$\neg$ (Negation) $\rightarrow \wedge$ (Conjunction) $\rightarrow \vee$

(Disjunction) $\rightarrow \Rightarrow$ (Implication) $\rightarrow \Leftrightarrow$

(Equivalence)

Use () to group clauses to specify order of priority.

- For example: $(A \wedge B \vee C) \rightarrow D \wedge E$



$A \vee B \wedge C$	$A \vee (B \wedge C)$
$A \wedge B \Rightarrow C \vee D$	$(A \wedge B) \Rightarrow (C \vee D)$
$A \Rightarrow B \vee C \Leftrightarrow D$	$(A \Rightarrow (B \vee C)) \Leftrightarrow D$

Propositional logic

Syntax & semantic

- The semantics of propositional logic allows us to determine the meaning of formulas in the real world.
 - A fact can only be true or false
 - For example: $P =$ "Hanoi is the capital of Vietnam"
-

Propositional logic

Syntax & semantic

Model (world / assignment) — interpretation

- A model m is a way of assigning the value **True/False** to every propositional symbol. Each model is "a possible world".
- If there are n symbols, there are 2^n models (2^n lines of the truth table).
- For example: with 2 symbols $P, Q \rightarrow$ there are 4 models:
 $m_1 = \{P=F, Q=F\}, m_2 = \{P=F, Q=T\}, m_3 = \{P=T, Q=F\}, m_4 = \{P=T, Q=T\}.$

What does "the sentence α is true in model m " mean?

- Assign values to symbols according to m , then calculate from inside to outside according to the **truth table** of each connective ($\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$) — obtain True or False.
- Notation: $m \models \alpha$ reads "m satisfies α " (α is true in m).
- $M(\alpha)$ = set of all models that make α true.

Propositional logic

Syntax & semantic

Model (world / assignment) — interpretation

For example, calculating the value for each model.

Consider the formula $\alpha = (P \vee Q) \wedge \neg Q$. Evaluate α across all four models:

Model	P	Q	$P \vee Q$	$\neg Q$	$\alpha = (P \vee Q) \wedge \neg Q$
m_1	F	F	F	T	F
m_2	F	T	T	F	F
m_3	T	F	T	T	T
m_4	T	T	T	F	F

$\rightarrow \alpha$ is true only in m_3 , i.e., $M(\alpha) = \{ m_3 \} = \{ P=T, Q=F \}$.

Propositional logic

Syntax & semantic

Entailment ($\models$) — "logically implies"

- $KB \models \alpha$: "KB **entails** α ," meaning: α is **true** in every model where KB is **true**.
- If one **accepts** KB, one is **compelled** to **accept** α .
- In terms of model sets: $KB \models \alpha \Leftrightarrow M(KB) \subseteq M(\alpha)$ (every world that satisfies KB also satisfies α).

How to check for entailment

- Consider all models; **only look at models as $KB = \text{True}$** ;
- If in **all** of them, α also = True $\rightarrow KB \models \alpha$.
- If there is even a model $KB=\text{True}$ but $\alpha=\text{False}$ $\rightarrow KB \not\models \alpha$.

Propositional logic

Syntax & semantic

Entailment ($\models$) — "logically implies"

Example: Given $KB = (P \Rightarrow Q) \wedge P$, does $KB \models Q$?

Model	P	Q	$P \Rightarrow Q$	$KB = (P \Rightarrow Q) \wedge P$	Q
m_1	F	F	T	F	F
m_2	F	T	T	F	T
m_3	T	F	F	F	F
m_4	T	T	T	T	T

- The KB **true** only at m_4 . At m_4 , $Q = \text{True} \Rightarrow KB \models Q$ ✓ (this is essentially Modus Ponens viewed from a semantic perspective).
- Conversely, $KB \not\models \neg Q$: because at m_4 (the only model satisfying the KB), $\neg Q = \text{False}$.

Propositional logic

Syntax & semantic

Soundness & Completeness

- $\models$ (semantics): $KB \models \alpha$ — " α is true in all models making KB true". This is the truth.
- $\vdash$ (syntax): $KB \vdash_i \alpha$ — " α is derived from KB by inference procedure i " (applying rules such as Modus Ponens, resolution...). This is the calculator that calculates it.

Does procedure $\vdash$ match truth $\models$?

Soundness

- If $KB \vdash \alpha$, then $KB \models \alpha$.
- Meaning: The procedure derives only what is actually true—it never yields an incorrect conclusion. ($\vdash \subseteq \models$)

Completeness

- If $KB \models \alpha$, then $KB \vdash \alpha$.
- Meaning: the procedure derives everything that is entailed—no truth is missed. ($\models \subseteq \vdash$)

"A procedure that is both sound and complete is $\vdash$ identical $\models$: it calculates correctly and has all logical consequences. That's what we want."

Propositional logic

Syntax & semantic

Soundness & Completeness

Example: SOUND Rule — Modus Ponens

From $P \Rightarrow Q$ and P , it follows that Q . ("It is raining" $\Rightarrow$ "the road is wet".)

Check if $KB = (P \Rightarrow Q) \wedge P \models Q$?

P	Q	$P \Rightarrow Q$	$KB = (P \Rightarrow Q) \wedge P$	Q
F	F	T	F	F
F	T	T	F	T
T	F	F	F	F
T	T	T	T	T

$\rightarrow$ The KB is true in only one row ($P=T, Q=T$), and in that row, $Q = T$. Thus, every model that satisfies the KB also satisfies $Q \rightarrow KB \models Q$. The rule is sound ✓.

Propositional logic

Syntax & semantic

Soundness & Completeness

Example: UNSOUND Rule — Affirming the Consequent

From $P \Rightarrow Q$ and Q , one might infer P . ("Wet road" $\Rightarrow$ "raining"?) — it sounds plausible but is incorrect: the road could be wet due to a street-watering truck, not rain.
Check if $KB = (P \Rightarrow Q) \wedge Q \models P$?

P	Q	$P \Rightarrow Q$	$KB = (P \Rightarrow Q) \wedge Q$	P
F	F	T	F	F
F	T	T	T	F $\leftarrow$ phản ví dụ
T	F	F	F	T
T	T	T	T	T

$\rightarrow$ There is a model $\{P=F, Q=T\}$ where the KB is true but P is False. Thus, $KB \not\models P \rightarrow$ the rule is NOT sound $\times$. That model corresponds to the scenario of "wet ground (Q) but no rain ($\neg P$)."

Propositional logic

Canonical Form

- Two propositions **A** and **B** are said to be **equivalent** ($A \equiv B$) if they have the same truth value (both true or both false) in all possible cases.

$$\begin{aligned}(\alpha \wedge \beta) &\equiv (\beta \wedge \alpha) && \text{commutativity of } \wedge \\(\alpha \vee \beta) &\equiv (\beta \vee \alpha) && \text{commutativity of } \vee \\((\alpha \wedge \beta) \wedge \gamma) &\equiv (\alpha \wedge (\beta \wedge \gamma)) && \text{associativity of } \wedge \\((\alpha \vee \beta) \vee \gamma) &\equiv (\alpha \vee (\beta \vee \gamma)) && \text{associativity of } \vee \\\neg(\neg\alpha) &\equiv \alpha && \text{double-negation elimination} \\(\alpha \Rightarrow \beta) &\equiv (\neg\beta \Rightarrow \neg\alpha) && \text{contraposition} \\(\alpha \Rightarrow \beta) &\equiv (\neg\alpha \vee \beta) && \text{implication elimination} \\(\alpha \Leftrightarrow \beta) &\equiv ((\alpha \Rightarrow \beta) \wedge (\beta \Rightarrow \alpha)) && \text{biconditional elimination} \\\neg(\alpha \wedge \beta) &\equiv (\neg\alpha \vee \neg\beta) && \text{De Morgan} \\\neg(\alpha \vee \beta) &\equiv (\neg\alpha \wedge \neg\beta) && \text{De Morgan} \\(\alpha \wedge (\beta \vee \gamma)) &\equiv ((\alpha \wedge \beta) \vee (\alpha \wedge \gamma)) && \text{distributivity of } \wedge \text{ over } \vee \\(\alpha \vee (\beta \wedge \gamma)) &\equiv ((\alpha \vee \beta) \wedge (\alpha \vee \gamma)) && \text{distributivity of } \vee \text{ over } \wedge\end{aligned}$$

Figure 7.11 Standard logical equivalences. The symbols α , β , and γ stand for arbitrary sentences of propositional logic.

Propositional logic

Canonical Form

- Using only (**AND** and **NOT**), or (**OR** and **NOT**), we can express **any logical expression**.
- **Conjunctive Normal Form (CNF)**
 - Is a conjunction (AND conjunction) of clauses
 - Each clause is an OR conjunction of simple propositional symbols
$$(p_1 \vee \dots \vee p_n) \wedge \dots \wedge (q_1 \vee \dots \vee q_m)$$
- **Disjunctive Normal Form (DNF)**
 - Is is an OR of clauses
 - Each clause is an AND of simple propositional symbols
$$(p_1 \wedge \dots \wedge p_n) \vee \dots \vee (q_1 \wedge \dots \wedge q_m)$$

Propositional logic

Canonical Form

Transform to CNF

$$B_{1,1} \leftrightarrow (P_{1,2} \vee P_{2,1})$$

Step 1: Eliminate Biconditional ($\leftrightarrow$)

- Rule: Replace $\alpha \leftrightarrow \beta$ with $(\alpha \rightarrow \beta) \wedge (\beta \rightarrow \alpha)$
- Result: $(B_{1,1} \rightarrow (P_{1,2} \vee P_{2,1})) \wedge ((P_{1,2} \vee P_{2,1}) \rightarrow B_{1,1})$

Step 2: Eliminate Implication ($\rightarrow$)

- Rule: Replace $\alpha \rightarrow \beta$ with $\neg\alpha \vee \beta$
- Result: $(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg(P_{1,2} \vee P_{2,1}) \vee B_{1,1})$

Step 3: Move Negation ($\neg$) Inward

Rule: Use De Morgan's laws and double negation

- $\neg(\alpha \vee \beta) = \neg\alpha \wedge \neg\beta$
- $\neg(\alpha \wedge \beta) = \neg\alpha \vee \neg\beta$
- $\neg\neg\alpha = \alpha$
- Result: $(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge ((\neg P_{1,2} \wedge \neg P_{2,1}) \vee B_{1,1})$

Step 4: Apply Distributive Law

Rule: Distribute $\vee$ over $\wedge$

- $\alpha \vee (\beta \wedge \gamma) = (\alpha \vee \beta) \wedge (\alpha \vee \gamma)$

Result: $(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1})$

Final CNF Form: $(\neg B_{1,1} \vee P_{1,2} \vee P_{2,1}) \wedge (\neg P_{1,2} \vee B_{1,1}) \wedge (\neg P_{2,1} \vee B_{1,1})$

Propositional logic

Canonical Form

Transform to CNF

$$(A \vee B) \rightarrow (C \rightarrow D)$$

1. Loại bỏ phép suy ra

$$\neg(A \vee B) \vee (\neg C \vee D)$$

2. Chuyển phủ định vào trong ngoặc

$$(\neg A \wedge \neg B) \vee (\neg C \vee D)$$

3. Phân phối

$$(\neg A \vee \neg C \vee D) \wedge (\neg B \vee \neg C \vee D)$$

Propositional logic

Canonical Form

Transform to CNF

Excisers: Convert these functions to CNF

1. $P \vee (\neg P \wedge Q \wedge R)$

2. $(\neg P \wedge Q) \vee (P \wedge \neg Q)$

3. $\neg(P \Rightarrow Q) \vee (P \vee Q)$

4. $(P \Rightarrow Q) \Rightarrow R$

5. $(P \Rightarrow (Q \Rightarrow R)) \Rightarrow ((P \wedge S) \Rightarrow R)$

6. $(P \wedge (Q \Rightarrow R)) \Rightarrow S$

7. $P \wedge Q \Rightarrow R \wedge S$

Thank you!